

G.1/04 (OJ 2006, 334) concerned **diagnostic methods**. The Enlarged Board held that a diagnostic method within the meaning of Art. 52(4) EPC 1973 had an inherent and inescapable multi-step nature. If diagnosis as the deductive medical or veterinary decision phase was a purely intellectual exercise, the feature pertaining to the diagnosis for curative purposes and the features relating to the preceding steps which were constitutive for making the diagnosis represented the essential features of a diagnostic method within the meaning of Art. 52(4) EPC 1973. Thus, in order to satisfy the requirements of Art. 84 EPC 1973, an independent claim relating to such a method must include these features. The Enlarged Board further held that, while essential features were for the most part of a technical nature, a non-technical feature constitutive for defining the invention must likewise be included as an essential feature in the independent claim. Thus, although diagnosis *stricto sensu* was a purely intellectual exercise unless it was carried out by a device, the feature pertaining to it was such an essential feature as to be included in the independent claim. See also chapter I.B.4.6.1 c) "Clarity of a claim relating to diagnostic methods".

G.1/07 (OJ 2011, 134) concerned **surgical methods**. The Enlarged Board referred to **G.1/04** and stated that a claim should explicitly specify all essential features and must be clear. Under Art. 84 EPC, whether or not a step being or encompassing a surgical step excluded from patentability can be omitted either by using positive wording for such omission like "pre-delivered" or by simply leaving it out from the claim depends on whether the claimed invention is fully and completely defined by the features of the claim without that step. See also chapter I.B.4.4.4 c) "Surgical step part of claimed method or simply a preparatory measure?".

In **T.2102/12** the application related to medical robotic systems which allow to robotically move a tool on an articulated arm in response to the surgeon's manipulation of an input device. The board referred to **G.1/07** and held that claiming the measurement of the movement of the tool while "unclaiming" the movement itself rendered the claim unclear under Art. 84 EPC. The claimed method of intertwined non-surgical ("claimed") and surgical ("unclaimed") steps was not comparable to that underlying **T.836/08**.

In decision **T.923/08** the board decided as follows: If a method for recording measurements on the human or animal body necessarily entails a surgical step in order to fix to the human or animal body a measuring device which is indispensable for performing the method, this step has to be deemed an essential feature of the method and is comprised in such a method, even if the claim contains no method feature explicitly directed to it. Such a method is excluded from patentability under Art. 53(c) EPC. The exclusion of such a surgical step, whether by stating that the surgically attached measuring device was already fixed to the body before the start of the method or whether by use of a disclaimer, breaches Art. 84 EPC 1973 because such a process claim does not then include all the essential features of the claimed invention.

3.3. Clarity of broad claims

It was established in **T.238/88** (OJ 1992, 709) that the clarity of a claim is not diminished by the mere breadth of a term of art (e.g. "alkyl") contained in it, if the meaning of such

term – either per se or in the light of the description – is unambiguous for a person skilled in the art (see also T 950/97, T 103/00, T 1173/03, T 2068/10, T 745/11; see also in this chapter II.A.6.).

However, in T 935/14 the board also pointed out that broad features could only be considered clear under the proviso that the borders of the – broad – scope of protection could be clearly inferred by the skilled person. This made the distinction between broad but clear and broad and vague.

In T 523/91 the board held that the wording "plastics in combination with inserts" and "this insert is combined with the plastics granules in a wound or folded manner" did not cause the subject-matter of the claim to be unclear within the meaning of Art. 84 EPC 1973. Although the nature of the combination covered many possibilities, the resulting subject-matter of the claim was broad rather than unclear. The broadness of a claim could not be contested on its own but only in conjunction with other criteria such as novelty, inventive step or reproducibility.

Likewise in T 688/91 the board stated that a broad claim was not to be equated with one lacking clarity. The claim was worded as follows: "Integrated microprogrammed apparatus [...] with apparatus to generate [...] T-states [...] characterised in that [...] the generation [...] of the [...] T-states [...] is in each case effected as a function of preceding T-states and of state parameters such as the instantaneous values of state and/or mode signals [...]". The board held that, even if the words "such as" were understood to mean "for example", with the result that the subsequent information was not to be interpreted in a restrictive sense, the words "state parameters" were clear in as much as they conveyed the fact that T-state generation was to be effected as a function of any parameters representing one of the states of the apparatus. Although the claim was to be regarded as broad, since the terms used were open to general interpretation, breadth was not to be equated with an absence of clarity.

In T 630/93 the board noted that Art. 84, first sentence, EPC 1973 implied that it was not always necessary for a claim to identify technical features or steps in full detail. The function of the essential features, although normally expressed in technical terms, was often to define the borders of an invention rather than details of the invention within those borders. Thus, essential features could often be of a very general character and in extreme cases indicate only principles or a new idea (see also chapter II.C. "Sufficiency of disclosure").

In T 29/05 it was held that the existence of unrelated subject matter in the claims at issue was merely hypothetical, and no evidence had been provided by the examining division to support its possible presence in the prior art. In the absence of such evidence, each and every nucleic acid molecule (as specified) had to be assumed to be related to the subject matter disclosed in the application. The fact that a large number of possible nucleic acid sequences might fall within the scope of the claim was not a reason for raising a lack of clarity objection.

In T.2154/11 the board held that the objections of the examining division that the expressions "metadata describing an event", "taking", "connecting" were broad and embraced well-known activities were not considered per se a valid reason for a lack of clarity. The fact that some of these method steps might be understood to refer to "mental acts" or "normal human actions" was also no valid reason for a finding of lack of clarity as long as these steps did not give rise to ambiguity, as in the case in hand. If a method step could be interpreted to refer to a normal human activity, then this assessment should be taken into account for novelty and inventive step.

The principle whereby a broad claim is not unclear per se has been applied in e.g. T.456/91, T.393/91, T.530/94, T.950/97, T.1345/08, T.2220/09, T.125/15, T.2676/16, T.901/16.

3.4. Functional features

In T.68/85 (OJ 1987, 228) the board found that functional features defining a technical result were permissible in a claim (i) if, from an objective viewpoint, such features could not otherwise be defined more precisely without restricting the scope of the invention, and (ii) if these features provided instructions which were sufficiently clear for the expert to reduce them to practice without undue burden, if necessary with reasonable experiments. The board further pointed out that the effort to define a feature in functional terms had to stop short where it jeopardised the clarity of a claim as required by Art. 84 EPC 1973. The following decisions agreed with these findings: T.139/85, T.292/85 (OJ 1989, 275), T.293/85, T.299/86 (OJ 1988, 88), T.322/87, T.418/89 (OJ 1993, 20), T.707/89, T.204/90, T.752/90, T.388/91, T.391/91, T.810/91, T.822/91, T.894/91, T.281/92, T.490/94, T.181/96, T.750/96, T.265/97, T.568/97, T.484/98, T.1186/01, T.295/02, T.499/02, T.1173/03, T.404/06, T.959/08, T.560/09, T.556/11, T.754/13, T.2067/12, T.2427/13, T.2114/14. Some of these decisions (see e.g. T.204/90, T.181/96, T.265/97) examine a third criterion, which is not strictly speaking a requirement under Art. 84 EPC 1973 and requires that (iii) the state of the art does not stand in the way of using such functional and therefore general and broad terminology.

In T.361/88 the board distinguished between two types of functional feature: the first type of functional feature is related to process steps which are known to the man skilled in the art and may easily be performed in order to obtain the desired result; the second type of functional feature consists of process steps defined by the result which is aimed at. This is also allowable as long as the man skilled in the art knows, without exceeding his normal skills and knowledge, what he has to do in order to obtain said result.

In case T.720/92 the board noted that the term "clarity" in Art. 84 EPC 1973 referred to the practical meaning of the language of the patent claims. Claims with functional features which do not enable the skilled person to carry out the invention in the light of the disclosure and on the basis of common general knowledge do not meet the requirement of clarity according to Art. 84 EPC 1973. A claim combining functional definitions limited to features, which a skilled person would have no difficulty in determining on the basis of common general knowledge, and a structural definition of the essential contribution of the applicant is not objectionable under Art. 84 EPC 1973.

In T. 560/09 the board emphasised with regard to technical features expressed in general functional terms that the function must be able to be verified by tests or procedures adequately specified in the description or known to the skilled person. That meant not only that a feature in the claim must be comprehensible, but also non-ambiguous in that it could be determined without any ambiguity whether the claimed functional requirement was satisfied (see T. 2427/13). Hence, means of distinction were mandatory in order to allow a definition by a function instead of by a structure in a claim.

In T. 243/91 it was stated that a functional feature was allowable if that feature provided a clear instruction to a skilled person to reduce it to practice without undue burden. In T. 893/90 the feature "being present in amounts and proportions just sufficient to arrest bleeding" was held to be a functional feature which defined a technical result which also constituted a testable criterion to be satisfied by the claimed pharmaceutical composition. Because such testing involved only routine trials, the adopted functional language was allowable. The introduction of a reference to specific amounts and/or proportions of the components would limit the claim and was not necessary. The situation in T. 893/90 was distinguished from the one before the board in T. 181/96. Although in the former case the testing might appear prima facie bothersome, it was nothing out of the ordinary for the field of medicine, involving only routine trials. In the case in hand, however, which concerned an apparatus for hydrostatically testing a sealing element of a threaded connection between two connected sections of pipe, there was no general type of pipe connections with generally well-defined ranges of dimensions which were thus generally available for verification of the functional features as such. In T. 1802/12 the board stated that the description did not contain any test to verify whether the functional features were present after having applied the claimed method. The board concluded that – contrary to the situation in T. 893/90, where a test was available – the requirements of Art. 84 EPC were not met.

In T. 391/91 claim 1 set out in general terms the sequence of steps to be followed in order to put the invention into practice, i.e. in order to produce unicellular micro-organism host cells having INA (ice nucleation activity) or enhanced INA. The claim was in fact a generalisation from the particular examples. The board stated that as there was no reason to doubt that it was possible to generalise the specific teaching of the examples given, it would be unfair to the appellant to require a restriction of the claim by incorporation therein of the specific features of the examples. The skilled person could use any suitable variant capable of providing the same effect of the invention. This might be tedious, but it was nothing out of the ordinary in this field and involved only routine trials. The claim was thus allowable under Art. 84 EPC 1973.

In T. 241/95 (OJ 2001, 103) the Swiss-type second medical use claim defined the disease or disorder to be treated with substance X as "condition which is capable of being improved or prevented by selective occupation of the serotonin receptor". The board held that this functional definition was unclear because no test was at hand to determine whether the therapeutic effects were a result of the newly discovered property of X of occupying the serotonin receptor or any other known or unknown property of that substance. See also T. 2321/13.

The board in T. 830/08 was concerned with the clarity of a claim directed to a second medical use. Such claims are considered clear only if the disease to be treated are clearly defined in it (e.g. T. 1048/98). In the case in hand the disease to be treated was defined in functional terms as "the preferential induction of apoptosis in a first population of cells compared to a second population of cells wherein the cells of the first population are tumour cells". The question was whether the skilled person could clearly attribute a disease or group of diseases to this functional definition. In the board's view, this was not the case. The skilled person reading this definition in claim 3 would be struck, on the one hand, by the explicit mention and the specific definition of the first population of cells and by the explicit mention, but absence of specific definition of the second population of cells, on the other hand.

In T. 1074/00 the board considered that the term "capable of hybridising under stringent conditions" was sufficiently clear for the purposes of Art. 84 EPC 1973, having regard to the particular nature of the subject-matter (see also T. 29/05). The board found that, although different experimental protocols might be applied for assessing hybridization under stringent conditions, this did not mean that these protocols would lead to different results as far as the detected nucleotide sequence was concerned. Moreover, it had to be taken into account that the present claim defined its subject-matter also by a further functional feature relating to the biological activity.

In T. 151/01 the product claim at issue was intended to be restricted vis-à-vis the prior art embodiments by a functional feature, namely that the amount of the ingredient present in the composition must be a "therapeutic amount". The board had no doubt that the skilled person was perfectly able in most cases to decide whether a certain amount of a given non-steroidal anti-inflammatory agent had a therapeutic effect or not. However, it pointed out that, in order to establish the lower limit of the therapeutic amount for a given non-steroidal anti-inflammatory agent, in other words, in order to clearly establish the scope of protection of the claims, a standard test was required, since the result would strongly depend on the experimental method used. As there was no such test in the description or known to the skilled person, the board concluded that the claim did not fulfil the requirement of Art. 84 EPC 1973.

In T. 143/06 the density of the product which was "close to the theoretical maximum" was identified by the board as a functional feature of the product inextricably linked to the process conditions for obtaining the product. There was thus no objection of lack of clarity under Art. 84 EPC.

3.5. Characterisation of an invention by a parameter

In decision T. 94/82 (OJ 1984, 75) the board ruled that the requirement of clarity could be fulfilled in a claim to a product if the characteristics of the product were specified by parameters related to the physical structure of the product, provided that those parameters could be clearly and reliably determined by objective procedures which were usual in the art (see also T. 452/91, T. 541/97, T. 437/98, T. 193/01). It is not mandatory to give instructions in the claim itself as to how the product is to be obtained (T. 94/82).